

TouchVR: A Modality for Instant VR Experience

Sungjae Cho
sungjaecho@postech.ac.kr
Dept. of CSE
POSTECH
Pohang, South Korea

Jungeun Lee
jelee@postech.ac.kr
Dept. of CSE
POSTECH
Pohang, South Korea

Inseok Hwang
i.hwang@postech.ac.kr
Dept. of CSE & Grad. School of AI
POSTECH
Pohang, South Korea

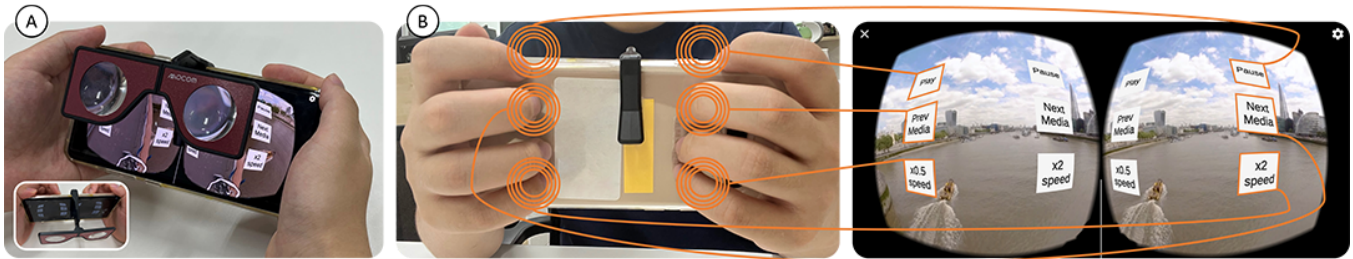


Figure 1: Using TouchVR in instant VR; (A) shows the setting of using the Pocket VR device. (B) shows 6 tap positions and mappings between positions and functions of our sample application.

ABSTRACT

In near future, we envision instant and ubiquitous access to the VR worlds. However, existing highly portable VR devices usually lack rich and convenient input modality. In response, we introduce TouchVR, a system that enables BoD interaction in instant VR supported by mobile HMDs.

We present the overall architecture and prototype of the TouchVR system in Android platform, capable of being easily integrated to Unity-based mobile VR applications. Furthermore, we implement a sample application (360° video player) to demonstrate the usage of our system. Through TouchVR, we expect it to be a step for enriching interaction methods in instant VR.

CCS CONCEPTS

• **Human-centered computing** → *Interaction techniques; Ubiquitous and mobile devices.*

KEYWORDS

Virtual Reality, Mobile Interaction, Inertial Sensor

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1 INTRODUCTION

Nowadays, numerous applications and enjoyable content (e.g., Roblox, VRChat) using VR are being introduced. Though most people currently use VR in dedicated purposes and time, we envision that they can pervasively access VR, anywhere and anytime – namely *Instant VR*. It is analogous to the advance of the Internet; people needed large and heavy desktops at first, but now it is instantly accessible through smartphones.

Head Mounted Displays (HMDs), such as Oculus Quest or HTC VIVE, have been introduced to help people enjoy VR in high quality and with rich input methods. However, they are heavy and hard to carry, thus unsuitable for instant VR. On the other hand, *mobile HMDs* have appeared, pursuing higher portability than *tethered* or *standalone HMDs*. The user may insert a smartphone into the housing (e.g., Google Cardboard) or clip the foldable lenses to a smartphone (e.g., Pocket VR in Figure 1(A)). They feature highly mobile-friendly form factors. However, they do not come with rich and convenient input methods, as their design is primarily focused on one-way information delivery – viewing. Current mobile HMDs mandate inconvenient and time-consuming input methods, e.g., users making a gaze to the Point-of-Interest (PoI) and then touching the screen or pushing a button [3, 4, 6] to interact with the PoI.

To overcome this limitation, we propose and implement TouchVR, a system that enables Back-of-Device (BoD) interaction in mobile HMDs, especially such as pocket VR where the user's both hands are occupied to hold the phone. The user can trigger different functions of the VR application by naturally tapping one of pre-defined regions at BoD with their fingers. Note that those pre-defined tapping regions are identical to the resting positions of the user's six fingers when naturally holding the phone as shown in Figure 1(B). This feature extends the input methods available to the user while not interfering with the natural ways to use mobile HMDs.

BoD interaction does not require any additional devices except the phone itself. Furthermore, BoD interaction in mobile phones are already validated in a number of works using built-in sensors, e.g.,

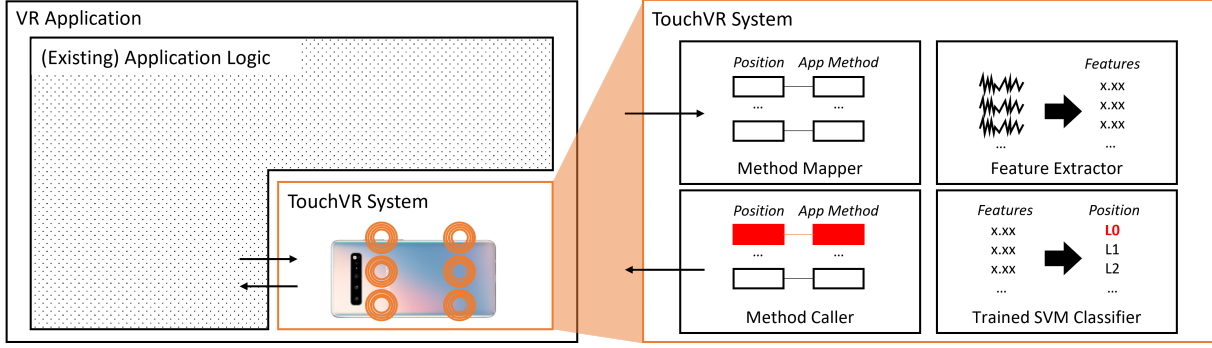


Figure 2: Architecture of TouchVR System

gyroscope, accelerometer, and microphone [5, 8], yet they have not been used for VR. We design our system based on BeyondTouch [8], but wrap them and provide APIs to make it easily integratable to mobile VR applications. With TouchVR, mobile VR applications can ascend to the level of richer two-way interaction, beyond the current mode which was mainly focused on one-way viewing.

2 SYSTEM ARCHITECTURE

Here, we describe each component of TouchVR and how they enable BoD interaction in mobile HMDs. The logical structure of TouchVR is shown in Figure 2.

Feature Extractor. Accelerometer and gyroscope sensor data are monitored for capturing the BoD tapping position. Feature Extractor computes 239 features from the sensor data within the time window, including time-domain features and frequency features used in [8]. **SVM Classifier.** A pre-trained classifier receives 239 features from the feature extractor and classifies a window as one class among 7 classes: L0 (top left), L1 (center left), L2 (bottom left), R0 (top right), R1 (center right), R2 (bottom right) and Other.

Method Mapper & Caller. Method Mapper and Caller enable other VR applications to take TouchVR inputs by mapping a tapping position to an existing method of the application, e.g., an input event handler. Method Mapper allows a VR application to register its method to a specific tapping position. Method Caller triggers the method registered for detected tapping position. The application developer may add lines to register or unregister their application’s method to a BoD tapping position to utilize TouchVR.

3 IMPLEMENTATION

3.1 TouchVR in Android

Referring to the architecture, we implemented TouchVR in Android platform. For hardware, we used a Samsung Galaxy S10. We used JTransforms [7] to perform Fast Fourier Transform (FFT) to extract frequency features. libSVM [1] was used for training, evaluating, and deploying the classifier. To collect a dataset to train the classifier, we developed a data collection application recording sensor data when tapping each BoD position.

3.2 Sample Application

In order to test the feasibility and real-time usage of our TouchVR system, we designed and implemented a simple 360° video player (*TouchVRPlayer*), which is easily accessible in mobile devices.

Our video player provides six functions: **Play, Pause, Previous video, Next video, 2X faster, 2X slower**. Each function is mapped to L0, R0, L1, R1, L2, R2 position, respectively. Figure 3 shows the appearance of the video player when the user tapped R0 position.

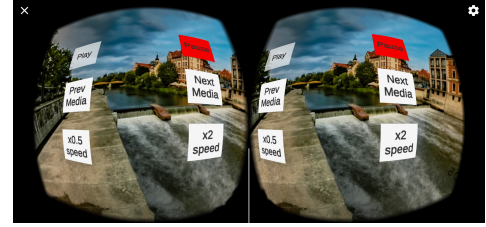


Figure 3: Sample application (360° video player)

Additionally, we implemented a baseline version of this application (namely *GazeVRPlayer*) with the same functions but using current input method, i.e., gaze-and-touch, to have a comparison with *TouchVRPlayer*.

4 DEMONSTRATION PLAN

In the demo, we will demonstrate both versions of the sample application: *TouchVRPlayer* and *GazeVRPlayer*. Devices for the demo are a smartphone (Samsung Galaxy S10) and a pocket VR device (MOCOM NABI). The user first grips a smartphone with their both hands and is requested to use *GazeVRPlayer* and explore naturally by using baseline input method. Then, we request the user to use our *TouchVRPlayer* by freely tapping 6 regions at BoD. With two versions, we expect to pull out comparative experiences between the baseline and the BoD input method in instant VR.

5 FUTURE USAGE OF TouchVR

Stock trade applications. Instant VR would excellently strengthen mobile stock trading application where a wide viewing area is essential to display multiple charts concurrently. As timely inputs are equally essential to make bidding or transactions without delay, the stock VR application developer can incorporate TouchVR to map functions, such as selling or buying, to BoD tapping positions. It will provide shortcuts to the user for triggering functions.

Visual chat applications. The immersive and extended viewing environment of Instant VR would innovate mobile multi-user visual chat experiences. By incorporating TouchVR, developers of such a chat application may map facial expressions (happy, sad, angry, etc.) or body reactions (clap, thumbs up, etc.) of an avatar to tapping positions. Eventually, instant VR enhanced by TouchVR would

spur the emergence of immersive social interaction in situated environment without necessitating extensive hardware setup [2].

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